MILITARY SPECIFICATION

CARTRIDGE, BOMB EJECTION CCU-10/B

This specification is mandatory for use by USAF

1. SCOPE

1.1 This specification covers the requirements for one type of cartridge designated as the Cartridge, Bomb Ejection CCU-10/B hereafter referred to as the cartridge. The cartridge is designed to provide a source of gas pressure for forced ejection of weapons from the Aircraft Bomb Ejector Racks MAU-12A/A, MAU-12B/A, MAU-12C/A, BRU-4/A, MAU-46/A, AERO-7/A, AERO-27/A and similar type racks.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-A-2550	Ammunition and Special Weapons, General Specification For
MIL-T-7743	Testing, Store Suspension Equipment, General Specification For
MIL-D-81303	Design and Evaluation of Cartridges for Stores Suspension Equipment

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronics and Electrical Parts
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

MIL-STD-831	Test Reports, Preparation of
MIL-STD-1167	Lot Numbering of Ammunition
MIL-STD-1168	Ammunition Data Card

DRAWINGS

Air Force

69E14040	Cartridge, Bomb Ejection CCU-10/B

OTHER PUBLICATIONS

Code of Federal Regulations

49CFR Parts 71-79	Interstate Commerce Commission Regulations for the Transportation of Explosives and other Dangerous Material

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Editions Revised 1965) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. Orders for the above publication should cite 49 CFR Parts 71-79 (Revised 1965).)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Design and Construction. The cartridge shall be constructed in conformance to Air Force drawing 69E14040 and subsidiary drawings.

3.3 Function. The function of the cartridge is to provide a source of gas pressure for forced ejection of weapons from Bomb Ejector Racks MAU-12B/A, MAU-46/A or similar types of ejector racks.

3.3.1 Ballistics. Two cartridges installed in a MAU-12 Bomb Ejector Rack and actuated by the "all fire" current shall produce performance as noted below. The test store shall be as described in 3.3.2.

	Velocity	Reaction Force
-65 ± 5°F	*	*
-70 ± 5°F	12.5 +2 fps separation -1 velocity	7600 lbs max Reaction Force
+325 ± 5°F	12.5 +2 fps separation -1 velocity	*

* Performance values at these temperatures are to be recorded for information only.

3.3.2 Store A. Store "A" shall have a weight of 500 ± 10 lbs, diameter of 14.75 ± 0.25 inches, and the center of gravity shall be 4.40 ± 0.25 inches forward of the aft lug. Fourteen-inch lug spacing shall be used.

3.3.3 Ejection velocities shall be measured at the end of the ejector rack piston stroke. Orifice sizes to be used are 0.110 inches diameter forward and 0.093 inches diameter aft.

3.4 Performance

3.4.1 All-Fire Sensitivity. The cartridge shall fire within 25 milliseconds, under all environmental conditions, when subjected to a direct current of 5 amperes at 24-28 volts d.c.

3.4.2 No Fire Sensitivity. The cartridge shall be capable of withstanding a minimum direct current of 1 ampere or 1 watt whichever is greater, for a duration of five minutes, minimum, without firing under any environmental conditions.

3.4.3 Sympathetic Ignition. The cartridge, when installed in the MAU-12 Rack shall fire sympathetically as a result of the electrically initiated firing, by the "all fire current" of a second cartridge.

3.4.4 Function Time. Two cartridges when installed in a MAU-12B/A rack which is loaded with the test store of para 3.3.2 shall exhibit a function time less than 100 ms. Paragraph 3.3.3 shall apply. Function time is defined as that time from the close of the firing key to the end of piston stroke.

3.5 Electrical Resistance. The electrical resistance of the ignition element shall be a minimum of 20 ohms after firing.

3.6 Environmental Requirements

3.6.1 Static Discharge. The cartridge shall not fire when subjected to an electrostatic discharge of 150,000 ergs from a 500 picofarad capacitor.

3.6.2 Stray Voltage. The cartridge shall be capable of proper performance after subjection to 2000 pulses of direct current. Each pulse shall be 105 ± 5 milliamperes. Duration of pulse shall be 300 milliseconds and pulse recurrent rate shall be 2.0 pulses per second.

3.6.3 Vibration. The cartridge shall be capable of satisfactory operation during and after vibration, as specified in vibration tests of this specification.

3.6.4 Air Tightness. The cartridge shall not exhibit a leak rate of greater than 1×10^{-4} cc/sec when subjected to a dry leak test at a pressure differential of 14 psi.

3.6.5 Temperature. The cartridge shall be capable of proper performance during and after prolonged exposure at temperatures from -65°F to +325°F.

3.6.6 Temperature Shock. The cartridge shall be capable of proper performance after being subjected to the temperature shock testing requirements of this specification.

3.6.7 Acceleration. The cartridge shall be capable of proper performance after sustaining a 50g acceleration for at least one minute in each of its three mutually perpendicular axes.

3.6.8 Acoustic Noise. The cartridge shall be capable of proper performance when subjected to acoustical noise levels of 160 decibels.

3.6.9 Impact Shock. The cartridge shall be capable of proper operation after being subjected to an 8-foot drop onto a steel plate supported by concrete.

3.6.10 Temperature and Humidity Cycling. The cartridge shall be capable of proper performance after being subjected to the temperature and humidity cycling test.

3.6.11 Protective Finish. The protective finish shall be in accordance with the applicable drawings and specifications.

3.7 Workmanship. The cartridge, including all parts and accessories, shall be fabricated in a manner that will ensure that the criteria of appearance, fit, and adherence to tolerances, as specified on the drawings, shall be observed. Particular attention shall be given to the markings required under "Identification of Product."

3.8 Identification of Product. The cartridge shall be marked for identification in accordance with MIL-STD-130.

3.9 Lot Numbering. Each separate production lot shall be assigned an ammunition lot number in accordance with MIL-STD-1167. This lot number shall be marked on each cartridge, as shown on the assembly drawing.

3.10 Storage Group. The cartridge belongs to storage compatibility Groups B, E, and N.

3.11 Quantity Distance Classification. The cartridge is a Class I Quantity-Distance item.

3.12 ICC Explosive Classification. CCU-10/B Cartridges shall be assigned ICC explosive classification in accordance with 49CFR Parts 71-79.

3.13 Government Loaned Property. The following items shall be loaned to the contractor, upon his request, as Government Loaned Property for the purpose of testing the cartridge:

Item Number	Part Number	Description of Item	Quantity
1	63J14360	Rack, Bomb Ejector, Aircraft MAU-12A/A	As Stated in Contract
2	64J13210	Rack, Bomb Ejector, Aircraft MAU-12B/A	As Stated in Contract
3	69J13060	Rack, Bomb Ejector, Aircraft MAU-12C/A	As Stated in Contract
4	68J13090	Rack, Bomb Ejector, Aircraft BRU-4/A	As Stated in Contract

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. The inspection and testing of the cartridge shall be as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 Preproduction Testing

4.2.1 Preproduction Testing. The preproduction test samples shall be representative of production cartridges. They shall be consecutively numbered for test identification. They shall be tested at a laboratory designated by the procuring activity, or when so stated in the contract at the contractor's plant under the supervision of the procuring activity. All samples shall be tested in the applicable bomb ejector rack.

4.2.2 Preproduction Test Report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods" as listed in Table I and Table II.

4.3 Acceptance Tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Plans and Tests

4.3.1 Individual Tests. Each cartridge shall be subjected to the individual tests listed in Table I.

4.3.2 Sampling Plan and Tests

4.3.2.1 Lot. Lot definition, size and formation shall be in accordance with MIL-STD-105 and shall contain:

a. Components produced by a homogenous process.

b. Propellant from not more than one lot.

c. Ignition composition from not more than one lot.

d. Ignition elements from not more than one lot.

4.3.2.2 Sampling Plan. The sample size shall be in accordance with MIL-STD-105, Inspection Level II and Table II-A, AQL of 1.0. Sample tests will be as stated in Table I of this specification. A sampling test plan shall be submitted to the procuring activity for approval before production begins.

4.3.3 Rejection and Retest. When a sample cartridge fails to meet specified test, no cartridge on hand or later produced shall be accepted until the extent and cause of failure are determined. The cause of failure shall be corrected and subsequent cartridges shall demonstrate compliance by passing the failed tests.

4.3.4 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions

wing the 1-hour flow period and shall be released from the nipple to

TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	PREPRO-DUCTION TEST	INDIVIDUAL TEST	SAMPLING TEST
Examination of Product	3.2	4.5.2	X	X	
Conformance to Drawings	3.2	4.5.3	X	X	
All Fire Sensitivity	3.4.1	4.5.5	X		X *
No Fire Sensitivity	3.4.2	4.5.6	X		X
Sympathetic Firing	3.4.3	4.5.7	X		
Electrical Resistance	3.5	4.5.8	X	X	
Static Discharge	3.6.1	4.6.1	X		
Stray Voltage	3.6.2	4.6.2	X		
Vibration	3.6.3	4.6.3	X		X
Airtightness	3.6.4	4.6.4	X		X
High Temperature	3.6.5	4.6.5	X		X *
Low Temperature	3.6.5	4.6.6	X		
Acceleration	3.6.7	4.6.8	X		
Acoustic Noise	3.6.8	4.6.9	X		
Temperature Shock	3.6.6	4.6.7	X		
Shock	3.6.9	4.6.10	X		
X-Ray		4.5.4	X		X
Ballistic Firing	3.3.1		X		X
Function Time	3.4.4		X		
Temperature Humidity Cycling	3.6.10	4.6.11	X		

* To be conducted in conjunction with ballistic firings.

TABLE II

PREPRODUCTION SCHEDULE AND TEST PLAN

TEST TITLE										
Air Tightness	1-192**									
X-Ray	1-192**									
Electrical Resistance	1-192**									
ENVIRONMENTAL TESTS										
Acoustic Noise		61-72							145-174	
Acceleration			73-84						145-174	
Temperature Shock				85-96					145-174	
Vibration					97-108					
Static Discharge						109-120			145-174	
Stray Voltage							121-132			
8 Ft Impact Shock								133-174		
Temp-Humidity Cycling									175-192	
PERFORMANCE FIRINGS										
No-Fire Sensitivity	1-192									
All Fire Sensitivity	1-192*									
Cartridge Firings -65°F	1-12	61-64	73-76	85-88	97-100	109-112	121-124	133-136	145-154	
Cartridge Firings +70°F	13-36	65-68	77-80	89-92	101-104	113-116	125-128	137-140	155-164,	175-192
Cartridge Firings +325°F	37-60	69-72	81-84	93-96	105-108	117-120	129-132	141-144	165-174	
Sympathetic Ignition	9-12	33-36	57-60	153-154	163-164	173-174				
Post Firing Electrical Resistance	1-192									
Function Time	1-192***									

*Except those units not subjected to current during sympathetic ignition.
**Tests will be repeated following the environment test phase.
***Shall not apply to sympathetic ignition.

4.4.1 Standard Conditions. Test conditions shall be as specified in each specific test under "4.5 Test Methods." If no conditions are stated, the following shall apply:

a. Temperature 21°C (70°F) ± 11°C (20°F).

b. Relative Humidity - 90 percent or less.

4.4.2 Test Temperature. The cartridge shall be stabilized at the specified temperature prior to conducting any tests. Temperature stabilization shall be as defined in MIL-STD-810.

4.4.3 Test Measurements

4.4.3.1 Instruments. All measurements shall be made with instruments whose accuracy has been verified. Calibration records shall be maintained.

4.4.3.2 Tolerances. The maximum allowable tolerance on test condition measurements shall be as follows:

a. Temperature - plus or minus 2.8°C (5°F).

b. Relative Humidity - plus or minus 5 percent.

c. Vibration and Shock Amplitude - plus or minus 5 percent.

d. Vibration Frequency - plus or minus 2 percent.

e. Current - plus or minus 2 percent.

f. Resistance - plus or minus 2 percent.

g. Time Interval - plus or minus 2 percent.

h. Pressure - plus or minus 5 percent.

i. Reaction Force - plus or minus 5 percent.

j. Velocity - plus or minus 5 percent.

k. Sound level - plus or minus 2 percent.

4.4.4 Required Measurements

a. Resistance of unit prior to firing.

b. Firing current.

c. Firing time.

d. Pressure vs time.

e. Reaction force vs time.

f. Pitch rate vs time.

g. Velocity.

h. For preproduction testing, the cartridge number shall be recorded.

i. Other measurements shall be taken as specified in preproduction schedule and test plan.

4.5 Test Methods

4.5.1 Classification of Tests. Inspection and testing of the cartridge shall be as shown in Table I.

4.5.2 Examination of Product

4.5.2.1 Examination of Components. One hundred percent examination shall be performed for all drawing characteristics when the component lot size is not in excess of 1000 cartridges. For lot sizes in excess of 1000 cartridges, sampling inspection will be performed utilizing a plan acceptable to the procuring agency, in accordance with MIL-STD-105. Failure to comply with all drawing and specification requirements shall be cause for rejection.

4.5.2.2 Examination of Assemblies. One hundred percent examination shall be performed to determine drawing compliance with all requirements of the assembly drawing and this specification when the assembly lot size is not in excess of 1000 cartridges. For lot sizes in excess of 1000, sampling inspection will be performed to determine drawing compliance utilizing a plan acceptable to the procuring agency in accordance with MIL-STD-105.

4.5.3 Conformance to Drawings. Each cartridge shall be inspected for strict conformance to the engineering drawings, and to specifications and standards referenced therein. Failure to meet these criteria shall be cause for rejection.

4.5.4 X-Ray Examination. Each cartridge of the preproduction test sample shall be subjected to X-ray examination. X-ray photographs shall be identified by date, cartridge part numbers, lot number and serial numbers. These X-ray photographs shall be examined to determine compliance with the requirements of this specification.

4.5.5 All Fire Sensitivity. Cartridge shall be tested according to Table II. Failure of any cartridge to fire, or failure to meet the ballistic requirements of this specification, shall be cause for rejection.

4.5.6 No Fire Sensitivity. Cartridges shall be tested according to Table II. The cartridges shall be held in a test jig. The direct current shall be 1 watt for a duration of five minutes. Firing of the cartridge shall be cause for rejection.

4.5.7 Sympathetic Ignition. A 5 ampere current shall be applied to one cartridge and the other shall be sympathetically ignited. The minimum velocity for the test store stated in the ballistic requirements of this specification do not apply to this test. Failure of the cartridge to achieve ignition shall be cause for rejection.

4.5.8 Electrical Resistance. The resistance of the bridge wire shall be measured within an accuracy of 1 percent using a test circuit which subjects the bridge wire to a current of 0.9 ± 0.05 amperes. The cartridge shall be subjected to this test current for a period not to exceed 60 seconds. This test shall be conducted in accordance with MIL-STD-202, Method 303.

4.6 Environmental Tests. During or after subjection to the environmental tests, as specified in the test, the cartridge shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental tests and cause for rejection.

4.6.1 Static Discharge. Twelve (12) cartridges shall be tested to the static discharge test and shall not fire when subjected to an electrostatic discharge of 150,000 ergs from a 500 picofarad capacitor.

4.6.2 Stray Voltage. Each of forty-two (42) cartridges shall be subjected to 2000 pulses of direct current. Each pulse shall be 100 milliamperes. Duration of pulse shall be 300 milliseconds and pulse recurrent rate shall be 2.0 pulses per second.

4.6.3 Vibration. Forty-two (42) cartridges shall be subjected to vibration tests prescribed in MIL-STD-810, Method 514, Procedure 1, except that Figure I and Table III of this document apply in lieu of Figure I and Table I in MIL-STD-810. Equal quantities shall be vibrated at -65°F, room temperature and +325°F after one hour soak time. Times shown in Table III refer to one axis of vibration.

RANGE CURVE FOR VIBRATION TEST

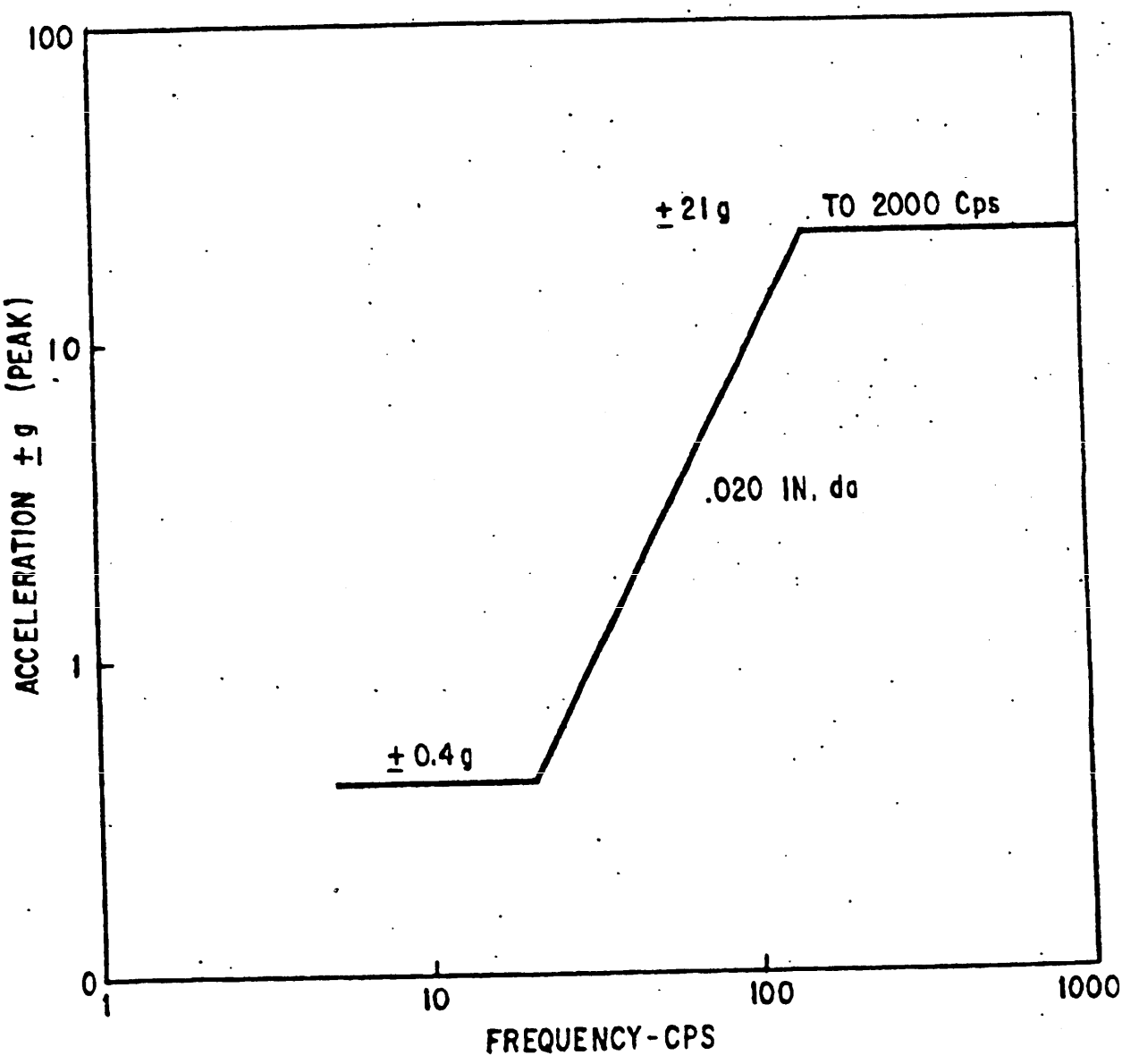


FIGURE 1

TABLE III

VIBRATION TEST SCHEDULE

NUMBER OF RESONANCES	TOTAL TIME AT RESONANCE	TOTAL CYCLE TIME
0		60 minutes
1	10 minutes	50 minutes
2	20 minutes	40 minutes
3	30 minutes	30 minutes
4	40 minutes	20 minutes

After the vibration test the 42 cartridges shall be surveyed and checked for electrical resistance and ballistically tested. Evidence of any disintegration of the ignition element, igniter or propellant

This constitutes one (1) cycle. The number of complete cycles shall be three (3). The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the cartridges shall be removed from the test chamber, conditioned, and tested according to Table I.

4.6.8 Acceleration. Forty-two (42) cartridges shall be subjected to tests in accordance with Method 513, Procedure I of MIL-STD-810, except that the cartridge shall be capable of withstanding a sustained acceleration of 50g for at least one minute.

4.6.9 Acoustic Noise. Forty-two (42) cartridges shall be subjected to tests in accordance with Method 515, Procedure I of MIL-STD-810, except for the using of an overall level of 160 db for the test level of Table II.

4.6.10 Impact Shock. Forty-two (42) cartridges shall be dropped from a height of eight feet onto a steel plate supported by concrete. Fourteen (14) cartridges shall impact nose down, fourteen (14) nose up, and fourteen (14) horizontally. Firing of a cartridge during this test shall be cause for rejection.

4.6.11 Temperature Conditioning. Ballistic firings will be conducted in accordance with Table II. Cartridges shall be soaked at the designated temperature for at least four hours. Cartridges shall be fired within three minutes after removed from the temperature chamber.

4.6.12 Temperature and Humidity Cycling. Eighteen (18) cartridges shall be tested in accordance with paragraph 3.4.5, MIL-D-81303. At the conclusion of the test the cartridges must meet the functional requirements of this specification and be free from visible damage.

4.7 Test Procedures shall be in accordance with MIL-T-7743.

4.7.1 Instrumentation. The instrumentation used to measure the current flow is to have an accuracy of $\pm$ 1 percent. When a rectified current is used the root mean square ripple content shall not exceed 5 percent of the test current and the rise time of the pulse shall not exceed 50 microseconds with a maximum overshoot of 5 percent.

4.7.2 Electrical Resistance. The resistance of the bridge wire shall be measured within an accuracy of 1 percent using a test circuit which subjects the bridge wire to a current of 0.9 $\pm$ 0.05 amperes. The cartridge shall be subjected to this test current for a period not to exceed 60 seconds. This test shall be conducted in accordance with MIL-STD-202, Method 303.

5. PREPARATION FOR DELIVERY

5.1 Data Cards. Data card information shall be as specified in MIL-STD-1167.

5.2 Packaging. For all levels of shipment, cartridges shall be packaged as specified by the procuring activity.

5.3 Marking. Exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Cartridge, Bomb Ejection CCU-10/B

5.3.1 Additional Marking (Explosive). In addition to marking in accordance with MIL-STD-129, the exterior containers shall be marked in accordance with Code of Federal Regulations 49CFR71-79. The following specific warning shall appear on the exterior container: "ELECTRIC EXPLOSIVE CARTRIDGES, Handle With Care."

6. NOTES

6.1 Intended Use. The Cartridge, Bomb Ejection CCU-10/B is intended for use in the MAU-12A/A, MAU-12B/A, BRU-2/A, BRU-4/A, MAU-46/A, AERO-7/A Bomb Ejector Racks or in similar applications where a source of high pressure gas generation is required.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaned property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procurement activity.

6.2.1 Preproduction Samples. It is expected that the contract or purchase order will specify that 192 cartridges, representative of the production cartridges will be submitted to the contractor or designated laboratory for test and evaluation. The invitation for bids, and the contract or purchase order, should specify the point of delivery for these items.

6.3 Production Delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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